Practice Quiz: Planning / Research Frontiers (Research Methods)

Name: __ ***Date:*** __

Part A: Multiple Choice

1. The most mature empirical application of Active Inference is: A) Robot control B) Dynamic Causal Modelling — with thousands of neuroimaging studies using its framework for effective connectivity inference C) Social media analysis D) Climate modeling

2. Active Inference's potential advantage over deep RL for AI safety is: A) It's slower B) Agents with prior preferences (C vector) pursue bounded objectives rather than unbounded reward maximization — potentially providing inherent safety constraints C) It requires more data D) It's less mathematical

3. The scale problem in Active Inference refers to: A) Model size B) Connecting free energy minimization at micro-levels (synapses, neurons) to macro-levels (behavior, societies) through formal multi-scale frameworks C) Hardware scaling D) Data volume

4. A strong Active Inference research proposal includes: A) Only theory B) Theoretical contribution, empirical predictions, detailed methods, preliminary data, and significance — bridging formal models with testable outcomes C) Only data D) Only code

5. The four tracks of the 401 course integrate by providing: A) Redundant information B) Philosophy (what and why) + neuroscience (where) + mathematics (how formally) + methods (how to test) — the complete toolkit for Active Inference research C) Only mathematical skill D) Only clinical training

6. After completing this course, a researcher should be able to: A) Only read papers B) Read/evaluate papers, design experiments, implement models, apply computational phenotyping, write proposals, and communicate findings — functioning as an independent Active Inference researcher C) Only run analyses D) Only write code

7. The FEP's application to morphogenesis suggests: A) Cells are conscious B) Biological development can be modeled as free energy minimization — cells differentiate and organize to minimize prediction error about their expected phenotype C) Only neural systems use FEP D) DNA directly encodes free energy

Part B: Capstone Essay Questions

1. Write a complete research proposal abstract (300 words) for an Active Inference study at one of the five frontiers (AI, consciousness, biology, ecology, scale). Include: background, objective, hypothesis, methods, expected results, significance.

2. Course capstone essay: Evaluate Active Inference as a scientific research program. Using Lakatos's framework of progressive vs. degenerative research programs, assess: (a) Does Active Inference generate novel predictions? (b) Have those predictions been confirmed? (c) Is the research program expanding or contracting? (d) What would constitute a "crucial experiment" for or against Active Inference? Cite specific examples from across the 401 course. (600 words)

3. Final synthesis: Imagine you are writing a review article titled "Active Inference at 20: Progress, Challenges, and Future Directions." Structure the review around: (a) the original vision (Friston 2006), (b) theoretical developments (variational calculus, information geometry, Bayesian mechanics), (c) empirical achievements (DCM, computational psychiatry, robotics), (d) philosophical contributions (enactivism, 4E cognition), (e) open challenges and future research priorities. This is your opportunity to demonstrate mastery of the entire 401 curriculum. (600 words)